Celsius, at which point it suddenly starts to freeze into solid ice. And, following the same reasoning regarding symmetries—but in reverse—we conclude that both of these phase transitions are accompanied by a *decrease* in symmetry.[23]

So much for ice, water, steam, and their symmetries. What does all this have to do with cosmology? Well, in the 1970s, physicists realized that not only can objects *in* the universe undergo phase transitions, *but the cosmosas a whole can do so as well.* Over the last 14 billion years, the universe has steadily expanded and decompressed. And just as a decompressing bicycle tire cools off, the temperature of the expanding universe has steadily dropped. During much of this decrease in temperature, not much happened. But there is reason to believe that when the universe passed through particular critical temperatures—the analogs of 100 degrees Celsius for steam and 0 degrees Celsius for water—it underwent radical change and experienced a drastic reduction in symmetry. Many physicists believe that we are now living in a "condensed" or "frozen" phase of the universe, one that is very different from earlier epochs. The cosmological phase transitions did not literally involve a gas condensing into a liquid, or a liquid freezing into a solid, although there are many qualitative similarities with these more familiar examples. Rather, the "substance" that condensed or froze when the universe cooled through particular temperatures is a field—more precisely, a *Higgs field.* Let's see what this means.

Force, Matter, and Higgs Fields

Fields provide the framework for much of modern physics. The electromagnetic field, discussed in Chapter 3, is perhaps the simplest and most widely appreciated of nature's fields. Living among radio and television broadcasts, cell phone communications, the sun's heat and light, we are all constantly awash in a sea of electromagnetic fields. Photons are the elementary constituents of electromagnetic fields and can be thought of as the microscopic transmitters of the electromagnetic force. When you see something, you can think of it in terms of a waving electromagnetic field entering your eye and stimulating your retina, or in terms of photon particles entering your eye and doing the same thing.